A 0D pulsatile model of aortic-arch banding

Reproducing carotid pulsatility redistribution (Eberth et al., 2009)

svZeroDSolver

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The problem

Eberth et al. (2009): a stiff band on the mouse aortic arch, placed *between* the two carotid takeoffs, creates paired carotids with similar mean pressure & flow but very different pulsatility.

- **RCCA-B** (right) — *upstream* of the band: sees the full pulse + reflections $\Rightarrow$ pulsatility index rises (PI ≈ 3.11).
- **LCCA-B** (left) — *downstream*: the resistive–inertial band damps the pulse (PI ≈ 1.65); baseline CCA ≈ 1.16.

Goal: a reduced-order (0D) lumped model that reproduces this redistribution with the band as the only structural change — and that maps to the wall growth & remodeling Eberth measured.

The 0D model

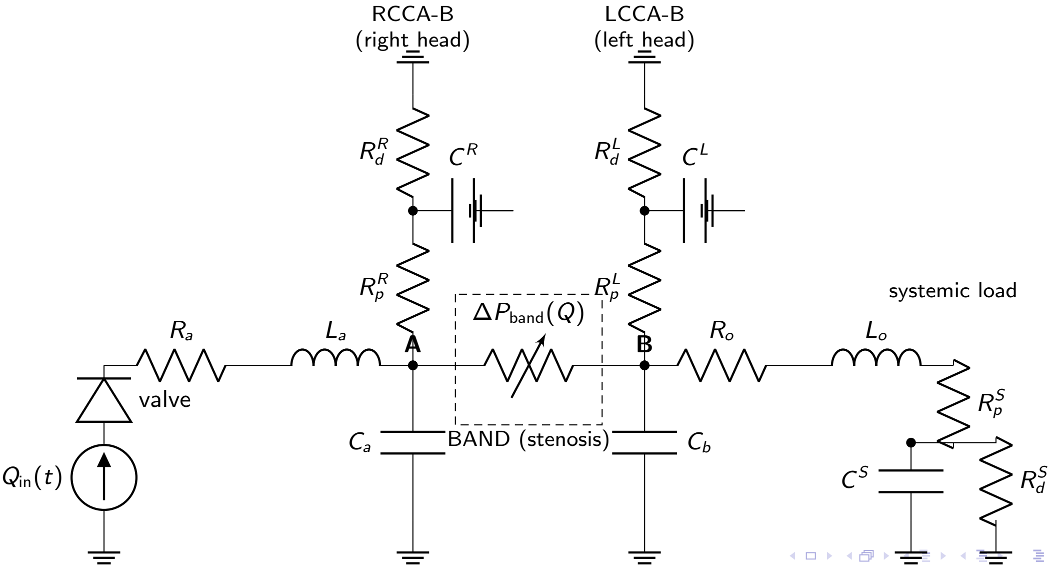
Heart $\to$ aortic valve $\to$ ascending aorta $\to$ **node A** $\to$ band $\Delta P(Q)$ $\to$ **node B** $\to$ descending aorta; a carotid RCR bed hangs off each node.

- Vessels: RCL `BloodVessel` blocks
- Band: nonlinear *Young–Tsai* stenosis $\Delta P = (R + S|Q|)Q + L\,\dot{Q}$
- Terminal beds: 3-element RCR windkessels
- Valve: `ValveTanh` diode

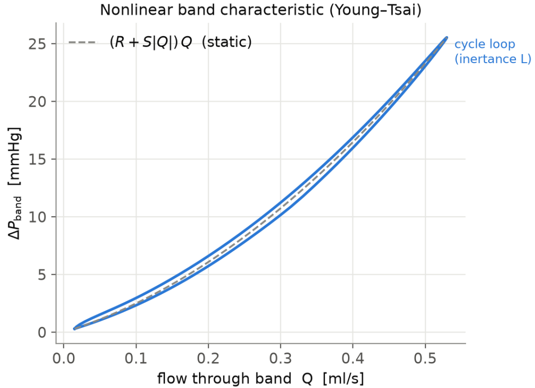
- Parameters from mouse literature (geometry, blood, wall stiffness E)
- Every parameter tagged FIXED / DERIVED / CALIBRATE
- Only the band `stenosis_coefficient` is calibrated (to the Table-1 PI ratio)

Three states differ *only* in carotid geometry: pre-banding (baseline), acute (baseline walls), chronic (Table-1 remodeled walls).

0D circuit (electric analogy)



The band is a nonlinear (Young–Tsai) element



ΔP–Q across the band: the quadratic Bernoulli/turbulent term ($S|Q|Q$) plus an inertance loop ($L\,\dot{Q}$). This nonlinearity + inertia is what damps the downstream pulse while leaving the upstream node pulsatile.

Parameter roles: fixed, assumed, and the knob we tune

Fixed (measured / literature)

- blood $\rho = 1.06$, $\mu = 3.5$ cP
- wall E: aorta 1.0, carotid 1.5 MPa
- HR 6.09 Hz, CO 0.20 ml/s, MAP 90
- carotid ID/wall (Table 1), aorta dia.
- measured central aortic compliance

$\Rightarrow$ all vessel R, L, C are *derived* from these via Eq. (1)–(3).

Assumed (plausible, not fitted)

- segment lengths (carotid, band) — estimates
- ejection waveform shape (systolic fraction)
- RCR split R_p:R_d = 10:90; time const. τ
- distal pressure $P_d = 0$
- valve R_{max}/R_{min}/steepness

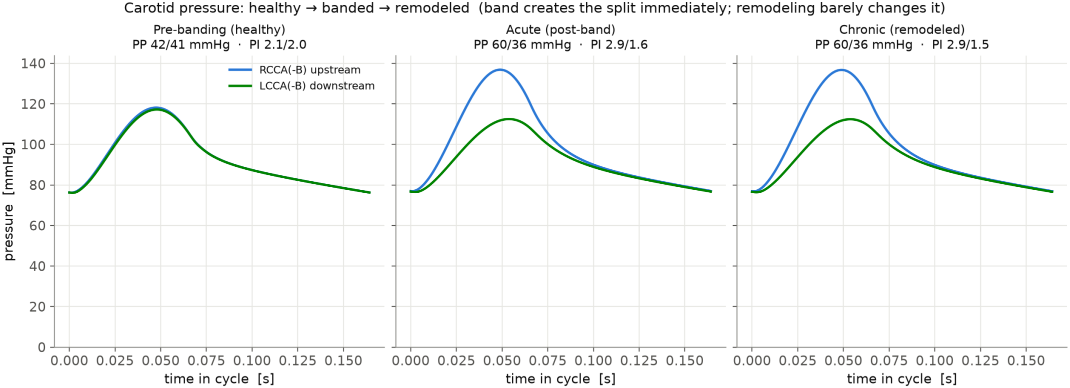
chosen by convention/estimate — could be refined, but not tuned to the data.

Knob (calibrated to data)

- band `stenosis_coefficient` S
- the **one** parameter we turn
- geometry gives a prior; tuned to the Table-1 PI ratio (3.11/1.65)

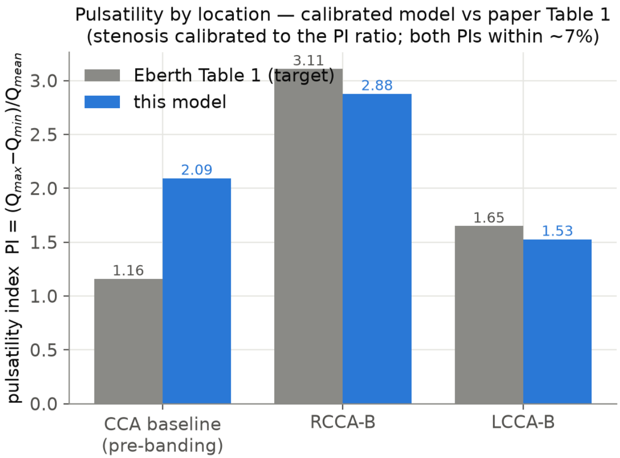
Everything else is fixed or derived — this is the only free knob fitted to the data.

Result: banding creates the split immediately



Carotid pressure is **symmetric before banding**, splits **immediately** on banding (upstream ↑, downstream damped), and **barely changes** with wall remodeling ⇒ the split is set by band *position*; remodeling is the response, not the cause.

Result: pulsatility vs. the paper



	Paper	Model
RCCA-B PI	3.11	≈ 2.9
LCCA-B PI	1.65	≈ 1.5
PI ratio	1.89	1.89
PP RCCA-B	56	60
MAP (R/L)	similar	98/91

The **banded** carotids, the PI ratio, MAP similarity, and pulse pressure are reproduced; mean carotid flows land in the measured range.

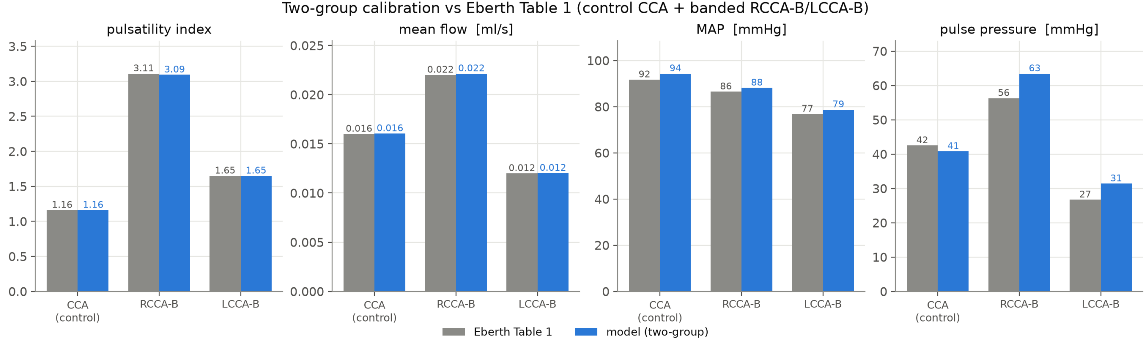
Limitation: baseline vs. banded cannot both be matched in 0D

- The model **over-predicts the baseline (pre-banding) PI** (≈ 2.1 vs. the paper's 1.16), though baseline *pulse pressure* matches (≈ 41 vs. 42).
- A parameter sweep shows every knob that lowers the baseline PI to 1.16 also collapses the banded PI — the passive bed is a linear pulsatility filter.
- The paper's baseline$\to$banded amplification is $\sim 2.7\times$ ($1.16 \to 3.11$); a lumped 0D band delivers only $\sim 1.5\times$.
- The missing $\sim 2\times$ is **wave reflection** off the band — a spatial/timing effect a 0D element cannot represent.

$\Rightarrow$ Reproducing the *full* amplitude needs a **1D** (wave-propagation) model — exactly the 0D$\to$1D boundary.

Revised: two-group calibration (control vs. banded)

Control (CCA) and banded (RCCA-B/LCCA-B) are **two separate animal groups** — they need not share $R/C/L$. Calibrating each on its own terms — HR and measured flows/pressures pin the beds, the band anchored to the measured A→B pressure drop, per-carotid bed compliance sets the PI — matches *all three* carotid states with a stock RCR:



DOF-balanced (8 targets / 8 knobs). Cost: bed compliance differs between groups (fitted, not measured) — descriptive, not a single predictive model.

Summary

- A literature-parameterized 0D model reproduces the Eberth **pulsatility redistribution**: upstream carotid pulsatile, downstream damped, at matched mean pressure/flow.
- Banding creates the split **immediately**; wall remodeling barely changes it — consistent with pulsatility being the *stimulus* for remodeling.
- With a *single shared* parameter set, 0D matches the redistribution and pressures but not both the baseline and banded PIs (a bounded amplification limit; a 1D wave-reflection effect is small for the mouse).
- Treating control and banded as *separate groups* — their own beds, band anchored to the measured pressure drop — matches **all three carotid states**, at the cost of group-specific (fitted) bed compliance.